Q1) CONCEPT MAPPING ITA0501

AUTONOMOUS VEHICLE

Uses cameras to identify roads, pedestrians, bicycles, traffic signals & obstacles while driving.

USES

OVERVIEW

Computer Vision enables machines to interpret and understand visual information from the real world.

provides intelligence to

COMPUTER VISION

works in

LEVELS OF COMPUTER VISION AND IMAGE PROCESSING

1. Low Level
2. Mid Level
3. High Level

follows a pipeline of

IMAGE SENSING AND ACQUISITION

- Cameras (RGB, stereo, depth)
- Capture real scenes
- Light to electrical

DIGITAL IMAGE FUNDAMENTALS

- 2D array of pixels
- Pixel value = intensity / colour
- Represented in grayscale or colour (RGB)

SAMPLING AND QUANTIZATION

- Sampling: Divide image into discrete pixels.
- Quantization: assign intensity values. (Eg: 8-bit, 256 levels.)

IMAGE FORMATION MODELS

- Pinhole camera model
- Perspective projective
- Imaging geometry and distortion.

FEATURE EXTRACTION & OBJECT RECOGNITION

- Detect edges, shapes, textures
- Identify objects (vehicles, signs)
- Use ML / Deep learning models.

- Amrita Anilkumar, 192471035

Q2) CONCEPT MAPPING ITA0501

MEDICAL IMAGING

Uses digital X-Ray images to help doctors identify abnormalities in patients through images, digitization & processing.

OVERVIEW

- X-rays are high energy electromagnetic waves that pass through the body.
- Detectors capture transmitted X-rays & convert to digital images.
- Images are processed & analysed using computer vision & image processing to assist doctors in diagnosis.

KEY FEATURES

- Non-invasive and safe
- High contrast for soft tissues
- Digital storage & easy sharing
- Enables image enhancement and automated analysis.
- Improves diagnostic accuracy and speed.

WORKFLOW

IMAGE SENSING AND ACQUISITION

- X Ray tube → X-Rays
- CCD
- Detector converts XRays to elec.

→

DIGITAL IMAGE FUNDAMENTALS

- 2D array pixels
- Grayscale
- Higher bit depth = better detail

→

SAMPLING & QUANTIZATION

- Converts into discrete pixels and assigns intensity levels

→

IMAGE FORMATION MODELS

- X-ray & tissue interaction
- Exponential attenuation
- Intensity distributions

→

LEVELS OF CV and IP

① Low level
② Mid Level
③ High Level

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